1 **(a)** $g(3) = 3^2 + 1 = 10$ **[M1]**

$f(10) = 2(10) - 5 = 15$ **[A1]**

(b) $y = 2x - 5$ OR $x = 2y - 5$ **[M1]**

$\frac{x+5}{2}$ **[A1]**

[Total Marks: 4]

2 **(a)** 10 **[B1]**

(b) 5 **[B1]**

(c) $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

[Total Marks: 4]

3 **(a)** **(i)** -14 **[B1]**

(ii) $f(2) = 3(2) - 2 = 4$ **[M1]**

20 **[A1]**

(b) $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

[Total Marks: 5]

4 **(a)** 18 **[B1]**

(b) $y = 5x - 2$ or $x = \frac{y+2}{5}$ **[M1]**

$\frac{x+2}{5}$ **[A1]**

(c) $g(2) = 2^2 + 3 = 7$ **[M1]**

$f(7) = 5(7) - 2 = 33$ **[A1]**

[Total Marks: 5]

5 **(a)** 10 **[B1]**

(b) $y = 3x - 2 \implies 3x = y + 2$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

(c) $f(2) = 3(2) - 2 = 4$ **[M1]**

$g(4) = 4^2 + 4 = 20$ **[A1]**

[Total Marks: 5]

6 **(a)** 13 **[B1]**

(b) Finds $f(2) = 1$ **[M1]**

-2 **[A1]**

(c) Sets $y = 5 - 2x$ and rearranges to $2x = 5 - y$ **[M1]**

$\dfrac{5-x}{2}$ **[A1]**

[Total Marks: 5]

7 **(a)** 5 **[B1]**

(b) $y = 4x - 3 \implies x = \dfrac{y+3}{4}$ or $x = 4y - 3$ **[M1]**

$\dfrac{x+3}{4}$ **[A1]**

(c) $f(1) = 4(1) - 3 = 1$ **[M1]**

$g(1) = 2 - 5(1) = -3$ **[A1]**

[Total Marks: 5]

8 **(a)** Calculates $g(3) = 2(3)^2 + 1 = 19$ **[B1]**

Substitutes their 19 into $f(x)$: $4(19) - 3 = 73$ **[B1]**

(b) Let $y = 4x - 3$ and attempts to rearrange for x (e.g., $y + 3 = 4x$) **[M1]**

$\dfrac{x+3}{4}$ **[A1]**

(c) Sets $4x - 3 = 17$ **[M1]**

$x = 5$ **[A1]**

[Total Marks: 6]

9 **(a)** 10 **[B1]**

(b) Sets $y = 3x - 2$ or $x = 3y - 2$ OR clearly shows inverse operations $(x + 2) \div 3$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

(c) Substitutes $f(x)$ into $g(x)$: $(3x - 2)^2 + 1$ **[M1]**

Correct expansion of $(3x - 2)^2$: $9x^2 - 6x - 6x + 4$ OR $9x^2 - 12x + 4$ **[M1]**

$9x^2 - 12x + 5$ **[A1]**

[Total Marks: 6]

10 **(a)** $y(x - 1) = 3x + 2$ **[M1]**

$xy - 3x = y + 2$ or $x(y - 3) = y + 2$ **[M1]**

$\dfrac{x+2}{x-3}$ **[A1]**

(b) $\dfrac{3x+2}{x-1} = x-2$ **[M1]**

$3x+2 = (x-1)(x-2)$ leading to $3x+2 = x^2-3x+2$ **[M1]**

$x^2-6x = 0$ or $x(x-6) = 0$ **[M1]**

0 and 6 **[A1]**

[Total Marks: 7]

11 (a) 1 **[B1]**

(b) $y = 2x+3 \implies 2x = y-3$ **[M1]**

$\dfrac{x-3}{2}$ **[A1]**

(c) $hk(x) = \dfrac{12}{(2x+3)-1}$ **[M1]**

$\dfrac{12}{2x+2} = \dfrac{6}{x+1}$ **[A1]**

(d) $k(2) = 2(2)+3 = 7$ **[B1]**

$\dfrac{12}{x-1} = 7 \implies 12 = 7(x-1)$ **[M1]**

$12 = 7x-7 \implies 19 = 7x \implies x = \dfrac{19}{7}$ (or $2\dfrac{5}{7}$) **[A1]**

[Total Marks: 8]

12 (a) Sets $y = \dfrac{4}{2x-3}$ and multiplies by denominator to obtain $y(2x-3) = 4$ **[M1]**

Expands and rearranges to collect terms in x: $2xy = 4+3y$ **[M1]**

$\dfrac{4+3x}{2x}$ **[A1]**

(b) Finds $g(-3) = 2$ **[B1]**

Equates and attempts to solve: $\dfrac{4}{2x-3} = 2 \implies 4x-6 = 4$ **[M1]**

2.5 (or $\dfrac{5}{2}$) **[A1]**

(c) Substitutes $f(x)$ into $g(x)$: $\dfrac{4}{2x-3}+5$ **[M1]**

Expresses over a common denominator: $\dfrac{4+5(2x-3)}{2x-3}$ **[M1]**

$\dfrac{10x-11}{2x-3}$ **[A1]**

[Total Marks: 9]

13 **(a)** $y = \dfrac{x+3}{x-1} \implies xy - y = x + 3$ **[M1]**

$xy - x = y + 3 \implies x(y-1) = y + 3$ **[M1]**

$\dfrac{x+3}{x-1}$ **[A1]**

(b) $\dfrac{\dfrac{x+3}{x-1} + 3}{\dfrac{x+3}{x-1} - 1}$ **[M1]**

$\dfrac{\dfrac{x+3+3(x-1)}{x-1}}{\dfrac{x+3-(x-1)}{x-1}}$ (or equivalent correct manipulation using a common denominator) **[M1]**

$\dfrac{x+3+3x-3}{x+3-x+1} = \dfrac{4x}{4} = x$ with no mathematical errors seen **[A1]**

(c) $f(3) = \dfrac{3+3}{3-1} = 3$ **[B1]**

$2x^2 - 5 = 3$ **[M1]**

$2x^2 = 8 \implies x^2 = 4$ **[M1]**

$2, -2$ **[A1]**

[Total Marks: 10]

14 **(a)** $y(x-1) = 3x + 2$ or $x(y-1) = 3y + 2$ **[M1]**

$xy - 3x = y + 2$ or $x(y-3) = y + 2$ **[M1]**

$\dfrac{x+2}{x-3}$ **[A1]**

(b) $fg(x) = \dfrac{3\left(\dfrac{2}{x+1}\right) + 2}{\dfrac{2}{x+1} - 1}$ **[M1]**

Numerator becomes $\dfrac{6+2(x+1)}{x+1}$ or Denominator becomes $\dfrac{2-(x+1)}{x+1}$ **[M1]**

$\dfrac{2x+8}{1-x}$ **[A1]**

(c) $\dfrac{3x+2}{x-1} = \dfrac{x+2}{x-3} \implies (3x+2)(x-3) = (x+2)(x-1)$ **[M1]**

$3x^2 - 7x - 6 = x^2 + x - 2$ **[M1]**

$2x^2 - 8x - 4 = 0$ or $x^2 - 4x - 2 = 0$ **[M1]**

$2 \pm \sqrt{6}$ **[A1]**

[Total Marks: 10]

15 **(a)** $y(x-1) = 3x + 5$ or $xy - y = 3x + 5$ **[M1]**

$xy - 3x = y + 5$ **[M1]**

$x(y-3) = y+5$ **[M1]**

$\dfrac{x+5}{x-3}$ **[A1]**

(b) $\dfrac{3x+5}{x-1} = x$ **[M1]**

$3x+5 = x^2 - x$ **[M1]**

$x^2 - 4x - 5 = 0$ **[M1]**

5 and -1 **[A1]**

(c) $f(2) = \dfrac{3(2)+5}{2-1} = 11$ **[M1]**

$11^2 + 1 = 122$ **[A1]**

[Total Marks: 10]

16 **(a)** $y = \dfrac{2x+1}{3} \implies 3y = 2x+1$ **[M1]**

$\dfrac{3x-1}{2}$ **[A1]**

(b) $hk(x) = \dfrac{2(5-x)+1}{3}$ **[M1]**

$\dfrac{11-2x}{3}$ **[A1]**

(c) $\dfrac{2x+1}{3} = 5 - x$ **[M1]**

$2x+1 = 15-3x \implies 5x = 14$ **[M1]**

2.8 (or $\dfrac{14}{5}$) **[A1]**

[Total Marks: 7]

17 **(a)** $h(5) = \dfrac{4}{5-3} = 2$ **[M1]**

$k(2) = 2(2)+1 = 5$ **[A1]**

(b) $y(x-3) = 4$ OR $x(y-3) = 4$ **[M1]**

$\dfrac{4}{x} + 3$ OR $\dfrac{4+3x}{x}$ **[A1]**

(c) $h(2x+1) = \dfrac{4}{(2x+1)-3}$ **[M1]**

$\dfrac{4}{2x-2} = 2 \implies 4 = 4x-4$ **[M1]**

$x = 2$ **[A1]**

[Total Marks: 7]

18 **(a)** $\dfrac{1}{5}$ **[B1]**

(b) $(2x+3)^2 - 4$ **[M1]**

$4x^2 + 6x + 6x + 9 - 4$ **[M1]**

$4x^2 + 12x + 5$ **[A1]**

(c) $y = 2x + 3 \implies y - 3 = 2x$ **[M1]**

$\dfrac{x-3}{2}$ **[A1]**

(d) 3 **[B1]**

[Total Marks: 7]

19 (a) 2 **[B1]**

(b) Sets $y = \dfrac{5}{x-2}$ and multiplies by $(x-2)$: $y(x-2) = 5$ **[M1]**

Rearranges to isolate x: $x - 2 = \dfrac{5}{y} \implies x = \dfrac{5}{y} + 2$ OR $xy - 2y = 5 \implies x = \dfrac{5+2y}{y}$ **[M1]**

$\dfrac{5}{x} + 2$ OR $\dfrac{5+2x}{x}$ **[A1]**

(c) Equates functions: $\dfrac{5}{x-2} = x + 2$ **[M1]**

Multiplies by $(x-2)$ to form a quadratic: $5 = (x+2)(x-2)$ **[M1]**

Expands and simplifies to $x^2 = 9$ or $x^2 - 9 = 0$ **[A1]**

3 and -3 **[A1]**

[Total Marks: 8]

20 (a) $(-2)^2 - (-2) - 7$ **[M1]**

-1 **[A1]**

(b) $(2x-3)^2 - (2x-3) - 7$ **[M1]**

$4x^2 - 12x + 9 - 2x + 3 - 7$ **[M1]**

$4x^2 - 14x + 5$ **[A1]**

(c) $2x - 3 = x^2 - x - 7$ **[M1]**

$x^2 - 3x - 4 = 0$ or $(x-4)(x+1) = 0$ **[M1]**

$x = 4$ and $x = -1$ **[A1]**

[Total Marks: 8]

21 (a) Substitution of $f(x)$ into $g(x)$: $\dfrac{5}{\dfrac{2x+1}{3} - 2}$ **[M1]**

Writing the denominator as a single fraction: $\dfrac{2x+1-6}{3}$ or $\dfrac{2x-5}{3}$ **[M1]**

$\dfrac{15}{2x-5}$ **[A1]**

(b) Let $y = \dfrac{5}{x-2}$ and attempting to multiply by $(x-2)$ or swapping x and y **[M1]**

$x - 2 = \dfrac{5}{y}$ or $xy - 2y = 5$ **[M1]**

$\dfrac{5}{x} + 2$ or $\dfrac{5+2x}{x}$ **[A1]**

(c) Evaluates $g(3) = 5$ **[B1]**

Equating $f(x)$ to their $g(3)$: $\dfrac{2x+1}{3} = 5$ **[M1]**

$x = 7$ **[A1]**

[Total Marks: 9]

22 (a) (i) 2.5 **[B1]**

(ii) $g(4-3x) = \dfrac{5}{(4-3x)-2}$ **[M1]**

$\dfrac{5}{2-3x}$ **[A1]**

(b) Minimum x-coordinate is $-\dfrac{-1}{2(1)} = 0.5$ or via completing square $(x-0.5)^2 - 7.25$ **[M1]**

$h(x) \geq -7.25$ **[A1]**

(c) $x^2 - x - 7 = 2x - 1$ leading to $x^2 - 3x - 6 = 0$ **[M1]**

$\dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(-6)}}{2(1)}$ **[M2]**

If M2 not scored, allow M1 for $\dfrac{3 \pm \sqrt{33}}{2}$ or correct use of formula with one error **[M1]**

-1.37 and 4.37 **[A1]**

[Total Marks: 9]

23 (a) $y(x-1) = 2x + 1 \implies xy - y = 2x + 1$ **[M1]**

$xy - 2x = y + 1 \implies x(y-2) = y + 1$ **[M1]**

$\dfrac{x+1}{x-2}$ **[A1]**

(b) $g(3x-2) = (3x-2)^2 - 4(3x-2)$ **[M1]**

$9x^2 - 12x + 4 - 12x + 8$ **[M1]**

$9x^2 - 24x + 12$ **[A1]**

(c) $3(x^2 - 4x) - 2 = x$ **[M1]**

$3x^2 - 13x - 2 = 0$ **[A1]**

$x = \dfrac{13 \pm \sqrt{(-13)^2 - 4(3)(-2)}}{2(3)}$ **[M1]**

$x = 4.48$ or $x = -0.15$ **[A1]**

[Total Marks: 10]

24 **(a)** Substitution of $g(x)$ into $f(x)$: $3(x^2 - x) + 2$ **[M1]**

$3x^2 - 3x + 2$ **[A1]**

(b) Substitution of $f(x)$ into $g(x)$: $(3x + 2)^2 - (3x + 2)$ **[M1]**

Correct expansion of $(3x + 2)^2$ to $9x^2 + 12x + 4$ **[M1]**

$9x^2 + 9x + 2$ **[A1]**

(c) Sets their three-term quadratic $gf(x)$ equal to 0: $9x^2 + 9x + 2 = 0$ **[M1]**

Valid method to solve quadratic (e.g., $(3x + 1)(3x + 2) = 0$ or correct use of formula) **[M1]**

$x = -\frac{1}{3}$ and $x = -\frac{2}{3}$ **[A1]**

(d) $5^x = 5^{-3}$ **[M1]**

$x = -3$ **[A1]**

[Total Marks: 10]

25 **(a)** Sets $y = \frac{2x + 5}{x - 3}$ and rearranges to get $y(x - 3) = 2x + 5$ **[M1]**

Expands and factorises to collect terms in x: $xy - 2x = 3y + 5 \implies x(y - 2) = 3y + 5$ **[M1]**

$\frac{3x + 5}{x - 2}$ **[A1]**

(b) Substitutes $g(x)$ into $f(x)$: $\frac{2(3x - 1) + 5}{(3x - 1) - 3}$ **[M1]**

Expands and simplifies the numerator to $6x + 3$ **[M1]**

$\frac{6x + 3}{3x - 4}$ **[A1]**

(c) Sets up the equation $3\left(\frac{2x + 5}{x - 3}\right) - 1 = x$ **[M1]**

Eliminates the denominator to obtain $6x + 15 - (x - 3) = x(x - 3)$ (or equivalent) **[M1]**

Simplifies to form the correct 3-term quadratic equation: $x^2 - 8x - 18 = 0$ **[A1]**

Correct use of the quadratic formula $x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(-18)}}{2(1)}$ **[M1]**

9.83 and -1.83 **[A1]**

[Total Marks: 11]

26 **(a)** $y = \frac{3x + 5}{x - 2} \implies y(x - 2) = 3x + 5$ **[M1]**

$xy - 2y = 3x + 5 \implies xy - 3x = 2y + 5 \implies x(y - 3) = 2y + 5$ **[M1]**

$\frac{2x + 5}{x - 3}$ **[A1]**

(b) $\frac{3}{(3x + 5)^2 + 2}$ **[M1]**

$\frac{3}{9x^2 + 30x + 25 + 2} = \frac{3}{9x^2 + 30x + 27}$ **[M1]**

$$\frac{3}{3(3x^2+10x+9)} = \frac{1}{3x^2+10x+9}$$ **[A1]**

(c) $\frac{3x+5}{x-2} = x \implies 3x+5 = x(x-2)$ **[M1]**

$3x+5 = x^2-2x \implies x^2-5x-5=0$ **[A1]**

$$\frac{-(-5) \pm \sqrt{(-5)^2-4(1)(-5)}}{2(1)}$$ **[M1]**

$$\frac{5 \pm \sqrt{25+20}}{2} = \frac{5 \pm \sqrt{45}}{2}$$ **[M1]**

5.85 and -0.85 **[A1]**

[Total Marks: 11]

27 (a) Sets $y = \frac{2x+3}{x-1}$ and multiplies: $y(x-1) = 2x+3$ **[M1]**

Expands bracket and collects x terms on one side: $xy - 2x = y + 3$ **[M1]**

Factorises x: $x(y-2) = y+3$ **[M1]**

$$\frac{x+3}{x-2}$$ **[A1]**

(b) Substitutes $f(x)$ into itself: $\dfrac{2\left(\frac{2x+3}{x-1}\right)+3}{\left(\frac{2x+3}{x-1}\right)-1}$ **[M1]**

Forms common denominators for numerator and denominator: $\dfrac{\frac{2(2x+3)+3(x-1)}{x-1}}{\frac{(2x+3)-(x-1)}{x-1}}$ **[M1]**

Correctly simplified numerator: $4x+6+3x-3 = 7x+3$ OR correctly simplified denominator: $2x+3-x+1 = x+4$ **[B1]**

$$\frac{7x+3}{x+4}$$ **[A1]**

(c) Calculates $f(2) = \frac{2(2)+3}{2-1} = 7$ **[B1]**

Forms correct quadratic equation: $x^2-4x-7=0$ **[B1]**

Correct substitution into quadratic formula: $\frac{4 \pm \sqrt{(-4)^2-4(1)(-7)}}{2}$ OR $\frac{4 \pm \sqrt{44}}{2}$ **[M1]**

5.32 and -1.32 **[A1]**

[Total Marks: 12]